Advanced Semi-Supervised Hyperspectral Change Detection via Cross-Temporal Spectral Reconstruction

Qinsen Liu and Bangyong Sun

***Abstract*—Hyperspectral change detection (HSI-CD) serves as an advanced technique for monitoring surface changes by leveraging spectral differences between multitemporal hyperspectral images (HSIs). Ideally, the spectral differences between unchanged areas across multitemporal images should be minimal. However, in practical applications, due to various factors such as imaging conditions and seasonal variations, spectral differences in unchanged areas can still be significant, which may lead to an increased likelihood of false alarms in CD. In this article, we propose a distinct reconstruction-guided approach for semi-supervised HSI-CD, termed cross-temporal spectral reconstruction network (CSR-Net), which can precisely distinguish the changed or unchanged areas with significant spectral differences. Specifically, CSR-Net encodes input HSI to reconstruct the corrected spectral sequences, mitigating errors caused by spectral differences in unchanged areas due to factors such as seasonal changes and variations in land cover characteristics. Subsequently, the reconstructed spectral sequences undergo change analysis to detect changes. Moreover, we propose an innovative semi-supervised HSI-CD loss, which weights the mean squared error loss based on binary CD results. This loss introduces a constraint that enables the model to learn spectral–temporal relationships from unlabeled data, thereby facilitating the reconstruction of corrected spectral sequences and reducing false alarms. Extensive experiments conducted on four benchmark HSI-CD datasets demonstrate that the proposed CSR-Net and loss function consistently outperform existing state-of-the-art methods, even with a very limited number of labeled samples.**

***Index Terms*—Change detection (CD), hyperspectral images (HSIs), spectral reconstruction, Transformer.**

I. Introduction

CHANGE detection (CD) can monitor the changes in the same area over different time periods through the analysis of multitemporal images [1]. It finds widespread applications in fields such as land management, urban planning, and natural disaster assessment.

With the continuous development of modern remote sensing imaging technologies, both the quality and quantity of remote sensing images have significantly improved compared with the past [2], [3], [4]. Hyperspectral imaging, as a cornerstone of modern remote sensing, provides unparalleled spectral resolution compared with multispectral or panchromatic systems, enabling precise characterization of material signatures. As one of the advanced methods of observing changes in the earth's surface, hyperspectral image CD (HSI-CD) can facilitate precise identification of changes by closely monitoring subtle spectral variations in the target area and has become a prominent research topic in the field of remote sensing [5].

Over the past decade, numerous HSI-CD methods have been proposed to detect changes between multitemporal images. In general, existing HSI-CD methods can be divided into two categories: traditional CD methods and innovative deep learning (DL)-based CD methods. Many traditional HSI-CD methods have been proposed, which can be divided into three categories: algebra-based methods, transformation-based methods, and classification-based methods. The algebra-based methods directly calculate the difference or ratio between multitemporal images to detect the changed areas. As one of the most commonly applied algebra-based CD methods [6], change vector analysis (CVA) detects change targets by calculating the change intensity and change direction of multitemporal images, but still has some difficulties, especially in selecting suitable thresholds. The transformation-based methods convert the multitemporal images into a unique feature space and utilize the feature maps to generate the CD result. Principal component analysis (PCA) [7], slow feature analysis (SFA) [8], and multivariate alteration detection (MAD) [9] are representative transformation-based CD methods, which have demonstrated promising detection performance in various applications. Especially, these transformation-based methods are effective for high-dimensional spectral processing and noise reduction, but the prominent time-consuming steps cannot be neglected during the calculation of the new component for large changed areas [10]. The classification-based methods compare classified multitemporal images pixel-by-pixel to realize CD. Support vector machine (SVM) [11] and extreme learning machine (ELM) are common models in

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the classification-based methods. These methods are able to deal with the heterogeneous images for CD, but they depend heavily on the performance of the classifier [12].

In recent years, DL methods have demonstrated superior performance in HSI-CD [13], [14], [15]. Based on learning strategies, DL-based HSI-CD methods can be categorized into fully supervised change detection (FSCD), unsupervised change detection (USCD), and semi-supervised change detection (SSCD). The task of FSCD can be considered as the full constraint to the reference map [16], [17]. With the learning of reliable labeled data, the FSCD methods have achieved outstanding CD performance [18], [19], [20]. However, it is hard to always provide an abundant training dataset in practical applications, since CD is a complex task and needs to interpret multitemporal images at the same time. USCD has no pregiven rules or labeled data for training, but there is prior knowledge that the representation of unchanged landscapes can be predicted from one image to another image, while the changed landscapes cannot. Based on the above theoretical foundations, some scholars have proposed some USCD algorithms [21]. Furthermore, SSCD uses a combination of both FSCD and USCD. The main idea of SSCD is to use both the distribution information of an abundant unlabeled sample and the reference map of a few labeled samples to improve the training of the network. With very few labeled data, SSCD can achieve rewarding CD performance. Many scholars have applied this training strategy to HSI-CDs [22].

With the rapid advancement of modern remote sensing technology, current hyperspectral imaging techniques now offer higher spectral resolution compared with the past, allowing for fine-grained classification of surface targets in the spectral dimension [23]. Most existing HSI-CD methods leverage the rich spectral information in multitemporal HSIs to deeply analyze the differences between these images, thereby facilitating CD [24]. For HSI-CD, target change areas are typically delineated based on human-defined criteria, covering changes in buildings, agricultural and forestry areas, and natural geological transformations. The spectral characteristics of these change areas often exhibit significant differences, and their change patterns are typically unpredictable. For the unchanged landscapes in multitemporal images, they have some consistency in spectral, spatial, or semantic aspects [1]. Ideally, unchanged areas in multitemporal HSI should consist of the same land cover type, exhibiting minimal spectral differences. However, seasonal impacts, shadowing, and other illumination variations can lead to a significant variance in the spectral change associated with unchanged areas, which may lead to false detections during CD processes [25]. These physical mechanisms are inherently predictable through modeling. Consequently, unchanged areas can be distinguished from real and false changes by modeling spectral–temporal relationships between multitemporal HSIs [26]. As shown in Fig. 1, in agricultural and forestry CD, farmlands in the same geographic area during spring and autumn should be classified as unchanged. However, variations in imaging conditions and seasonal transitions induce substantial spectral differences across temporal phases; these pseudo-changes thereby elevate false detection risks [27].

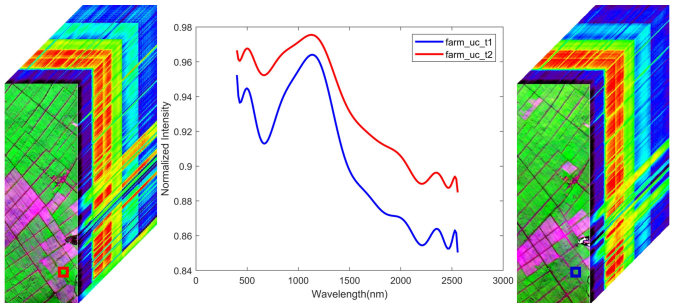


Fig. 1. Motivation. Due to factors such as imaging conditions and seasonal variations, significant spectral differences exist between unchanged areas in multitemporal HSIs (represented by the red and blue lines), which may lead to false detections in HSI-CD.

Based on the aforementioned motivations, we propose a distinct reconstruction-guided approach for semi-supervised HSI-CD, termed cross-temporal spectral reconstruction network (CSR-Net), which explicitly models spectral–temporal dependencies to reconstruct calibrated spectral sequences from multitemporal images. Unlike existing HSI-CD architectures that primarily focus on direct feature extraction or difference enhancement, CSR-Net introduces a dedicated spectral reconstruction stage to explicitly calibrate input sequences, thereby mitigating spectral variability. Subsequently, binary change detection (CD) is achieved through sequence difference analysis applied to the reconstructed spectral sequences, resulting in enhanced robustness against pseudo-changes. Furthermore, we introduce a novel semi-supervised HSI-CD loss that incorporates a constraint for selective gradient optimization. Distinct from existing loss functions, the proposed formulation adaptively weights the mean squared error (mse) loss using binary CD results as pseudo-labels. This design strengthens gradients in unchanged regions while suppressing those in changed areas, thereby providing more stable and targeted guidance during model training.

The main contributions of the proposed CSR-Net can be summarized as follows.

1) This article presents a distinct reconstruction-guided approach for semi-supervised HSI-CD, termed CSR-Net, which is capable of modeling spectral–temporal dependencies across multitemporal HSIs to reconstruct spectral sequences and subsequently analyzing these reconstructed spectral sequences for CD.
2) A novel semi-supervised HSI-CD loss function has been proposed, which incorporates an innovative constraint that adaptively strengthens gradients in unchanged regions while suppressing gradients in changed regions, thereby mitigating interference from false alarms caused by pseudo-changes.
3) The experimental results demonstrate that CSR-Net achieves state-of-the-art CD performance on four publicly available HSI-CD datasets using only a minimal amount of labeled data.

II. Related Work

The related works are considered in two groups: supervised and unsupervised HSI-CD methods, and semi-supervised HSI-CD methods.

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A. Supervised and Unsupervised HSI-CD Methods

In the past few years, many DL networks and their variants have been applied in HSI-CD. Since the reference map can provide the most intuitive supervised information, many scholars have proposed a variety of HSI-CD methods based on supervised learning and achieved excellent CD performance. For example, Sun et al. [28] performed deep feature encoding along the spectral dimension of HSIs to generate spectral tokens for CD. In addition, unsupervised learning does not require a reference map as supervisory information; it can learn from the statistical analysis of the internal features of unlabeled data. Li and Ding [29] developed a spectral–temporal Transformer network that utilizes self-attention mechanisms to capture complex spatial–spectral–temporal dependencies in bitemporal images, aiming to learn highly discriminative feature representations for CD. Similarly, Luo et al. [30] proposed a multiscale diff-changed feature fusion network to extract and enhance multiscale change features from differential representations. The core objective of these methods is to improve the network's ability to distinguish between changed and unchanged pixels. While such supervised HSI-CD methods demonstrate advantages in detection accuracy, they heavily rely on reference maps to provide stable gradients, posing certain limitations in practical applications. Most existing approaches implicitly extract spectral–temporal features through feature extraction frameworks, failing to explicitly address the issue of spectral variability.

Unsupervised learning does not require a reference map as supervisory information; it learns from the statistical analysis of internal features in unlabeled data. Therefore, many scholars have proposed a variety of unsupervised HSI-CD methods to address the issue that the reference map is difficult to obtain in practical applications. For example, Zheng et al. [31] innovatively proposed a cross-resolution difference learning method to realize unsupervised CD for different resolution images. In addition, numerous scholars have explicitly incorporated information from unchanged areas by extracting their features to enhance CD performance. Du et al. [32] proposed unsupervised deep SFA (DSFA), which specifically suppresses features in unchanged regions while highlighting change information. CFNet [33] introduced a content-aware mechanism to mitigate the impact of stylistic variations on features of unchanged areas, thereby suppressing interregion differences. Although existing unsupervised HSI-CD methods focus on invariant features in unchanged regions, their unsupervised loss functions struggle to provide stable gradient directions, making it challenging to achieve high CD accuracy.

In general, supervised and unsupervised learning strategies have been widely employed in HSI-CD. While each approach offers distinct advantages, they still face challenges such as insufficient accuracy or limited applicability. However, there is a pressing need to develop an explicit novel HSI-CD method that can effectively address the issue of spectral variability in multitemporal HSIs to achieve superior CD performance. Therefore, this article proposes the CSR-Net framework specifically to address this research objective.

B. Semi-Supervised HSI-CD Methods

As a combination of supervised and unsupervised learning, semi-supervised learning enables both feature learning from feature distributions in unlabeled data and learning supervisory information from much less labeled data. Compared with supervised learning, it leverages the feature distributions of abundant unlabeled data. Compared with unsupervised learning, it can learn supervised information from less labeled data. Therefore, many scholars have now applied the semi-supervised approaches to HSI-CD. For example, Liu et al. [34] developed a semi-supervised HSI-CD with a multilayer cascade screening strategy, which incorporates both spatial information and active learning to select highly reliable unlabeled data for increasing the training samples. Wang et al. [35] introduced a generative pretrained Transformer (GPSST) for semi-supervised HSI-CD, which incorporates a masked autoencoder pretrained on a large unlabeled HSI corpus to learn general spectral–spatial representations. A key drawback of this method is its requirement for large-scale external data during the pretraining phase. Huang et al. [36] introduced a novel cross-temporal bidirectional attention mechanism to capture bidirectional temporal dependencies by computing attention maps between tokens derived from bitemporal images. However, the loss function of its unsupervised branch relies on a predetermined threshold to generate pseudo-labels for gradient propagation, and the selection of this threshold significantly influences model accuracy. Furthermore, Hu et al. proposed SemiPSENet [37], which integrates a GELU_PSP module for multiscale feature extraction and employs an SE attention module to enhance feature representation, demonstrating effective utilization of both labeled and unlabeled data. However, SemiPSENet is primarily designed for and applied to red, green, blue (RGB) optical images, lacking the capability to process the rich spectral information inherent in hyperspectral imagery. Consequently, it fails to address the issue of spectral distortion in unchanged areas.

Overall, semi-supervised learning surpasses purely supervised or unsupervised methods by effectively combining limited labeled data with abundant unlabeled data. However, most existing SSCD methods overlook the spectral differences in unchanged areas across multitemporal HSIs, which often lead to false alarms. Crucially, since unchanged areas within the same region belong to identical land-cover categories, their spectral variations exhibit systematically modelable patterns. Thus, this article investigates a novel semi-supervised loss function that enables the model to adaptively enhance gradients in unchanged regions during the optimization process, thereby mitigating interference caused by false alarms from pseudo-changes.

III. Methodology

Compared with the other imaging technologies, HSIs offer rich spectral information, enabling strong endmember and material identifiability [38], where endmember refers to the standard term for a material's hyperspectral signature [39]. This capability enables HSI-CD to achieve fine-grained analysis through spectral signal decomposition across multitemporal

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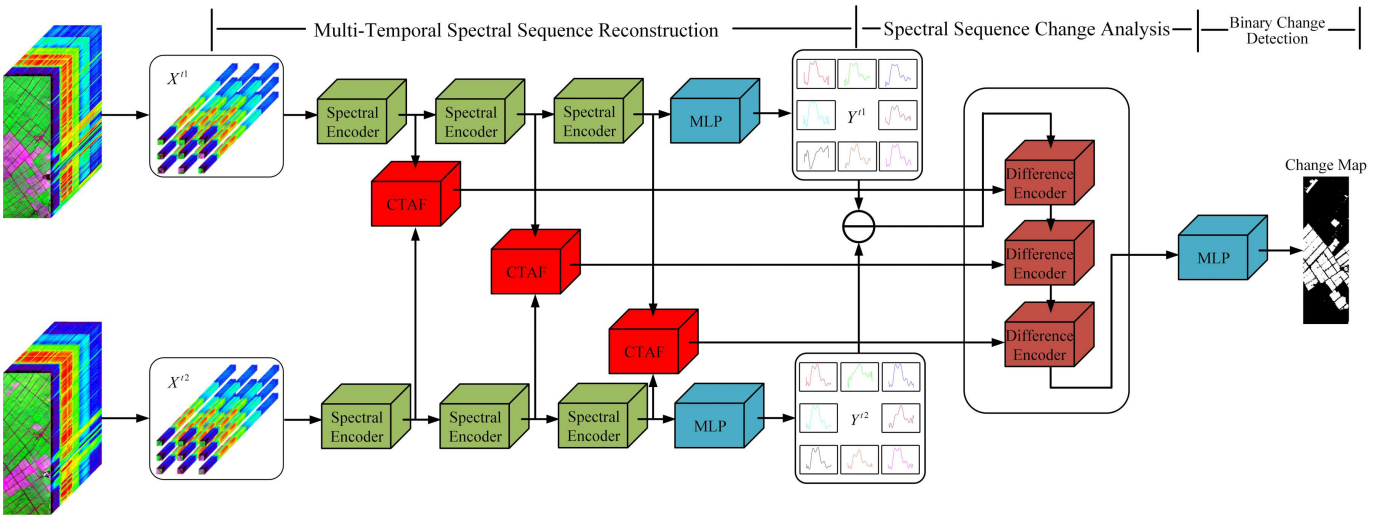


Fig. 2. Network structure of the proposed CSR-Net, which comprises three main components: cross-temporal spectral sequence reconstruction, spectral sequence change analysis, and binary CD. Initially, spectral sequences are reconstructed using spectral encoders, while the CTAF module outputs cross-temporal fused features. These reconstructed sequences and fused features are subsequently analyzed for changes through the difference encoders. Finally, an MLP processes the resulting difference sequences to generate binary classification results.

HSIs. However, while abundant spectral data improve detection capability, they also introduce confounding factors, such as spectral differences in unchanged areas due to variations in imaging conditions [27]. To effectively model the spectral variations of land-cover targets in multitemporal HSIs, we propose a distinct reconstruction-guided approach for semi-supervised HSI-CD, termed CSR-Net. The overall framework structure is shown in Fig. 2. The detection process comprises three main components: multitemporal spectral sequence reconstruction, spectral sequence change analysis, and binary CD. As shown in Fig. 2, multitemporal HSIs are first divided into patch pairs before being input to the framework. Within the first step of the framework, the HSI patches are encoded in spectralwise to reconstruct the corresponding spectral sequence. Second, the differences between the multitemporal reconstructed spectral sequences are encoded during change analysis. Finally, the change analysis result is fed into a multilayer perceptron (MLP) to detect changes. In this article, we use the Transformer as the basic feature encoder, given its ability to efficiently capture long-term spectral–temporal dependencies across multitemporal images [40].

A. Cross-Temporal Spectral Sequence Reconstruction

The changed areas between multitemporal HSIs are typically caused by factors such as natural geological changes and human activities, and the temporal relationships of these changes are unpredictable. In contrast, spectral differences in unchanged areas of multitemporal HSIs can be significant due to imaging conditions, seasonal variations, and land cover changes. Compared with the spectral differences between changed areas, the spectral–temporal dependencies of spectral variations between unchanged areas can be predicted.

Based on the aforementioned viewpoints, this section proposes a cross-temporal spectral reconstruction subnetwork, which is capable of modeling the spectral–temporal dependencies in multitemporal HSIs, reconstructing spectral sequences

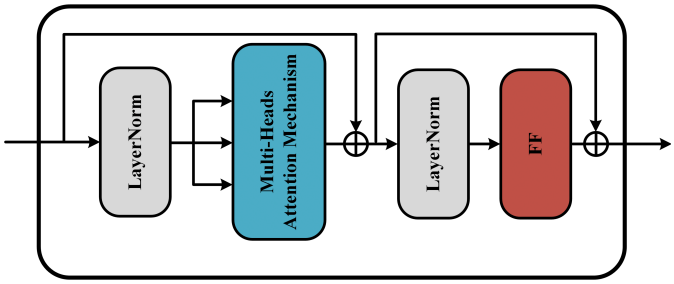


Fig. 3. Basic structure of the spectral encoder.

to suppress noise in unchanged areas, with the aim of reducing spectral differences in the unchanged areas. We first partition the multitemporal HSIs into overlapping patch pairs. Each patch $\boldsymbol{X}^{t1} \in \mathbb{R}^{c\times p\times p}$ corresponds to a spatial region of size $p \times p$ pixels, where c denotes the number of spectral channels. The patch is then transformed into a matrix $c \times p^2$ and uses a linear projection to map the sequence data into a set of d dimensions vectors, which can be formulated by

$$\boldsymbol{X} = \omega_1 \boldsymbol{X}^{t1} \tag{1}$$

where ω_1 is the trainable weight matrix with size of $p^2 \times d$, and $\boldsymbol{X} \in \mathbb{R}^{c\times d}$ is the feature embedding of temporal-one.

Considering that position embedding is important for capturing the order of an input sequence [41], we add position embedding $\mathrm{PE} \in \mathbb{R}^{c\times d}$ into $\boldsymbol{X}$, where PE is a randomly generated matrix. The updated embedding features can be expressed as

$$\boldsymbol{X}_0 = \boldsymbol{X} + PE. \tag{2}$$

After adding the position embedding, $\boldsymbol{X}_0 \in \mathbb{R}^{c\times d}$ are sent to the spectral encoders. As shown in Fig. 3, the spectral reconstruction encoder consists of alternating layers of multi-head self-attention (MHSA) mechanism and feed-forward (FF)

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blocks. MHSA is utilized to learn the multiple dependencies from the spectral sequence, which can be described as follows:

$$\mathrm{MHSA}(\boldsymbol{Q},\boldsymbol{K},\boldsymbol{V}) = \mathrm{Concat}(\mathrm{head}_1,\mathrm{head}_2,\ldots,\mathrm{head}_h)\,W^o$$
$$\mathrm{head}_i = \mathrm{softmax}\left(\frac{\boldsymbol{Q}_i\boldsymbol{K}_i^T}{\sqrt{d}}\right)V_i \tag{3}$$

where h is the number of heads, W^o is a learnable transformation parameter. $\boldsymbol{Q}_i = \boldsymbol{X}_i W_i^q$, $\boldsymbol{K}_i = \boldsymbol{X}_i W_i^k$, and $\boldsymbol{V}_i = \boldsymbol{X}_i W_i^v$ denote the query, key, and value, respectively, where W_i^q, W_i^k, W_i^v are three trainable matrix parameters and $\boldsymbol{X}_i \in \mathbb{R}^{c\times d}$ represents the feature embedding of the i layer. FF is also a key block in the Transformer structure, which holds two linear transformations with a Gaussian error linear unit (GELU) function. The FF can be formulated as follows:

$$\mathrm{FF}(x) = \mathrm{GELU}(\boldsymbol{x}W_1 + b_1)\,W_2 + b_2 \tag{4}$$

where W_1 and W_2 are parameter matrices; and b_1 and b_2 are biases, respectively. The features are updated as below

$$\boldsymbol{X}'_L = \mathrm{MHSA}(\mathrm{LN}(\boldsymbol{X}_{L-1})) + \boldsymbol{X}_{L-1}$$
$$\boldsymbol{X}_L = \mathrm{FF}\left(\mathrm{LN}\left(\boldsymbol{X}'_L\right)\right), L \in \{1,2,3\} \tag{5}$$

where LN(·) is the layer-norm. L is the number of Transformer encoder that belongs to $\{1,2,3\}$. Finally, an MLP is used to infer the encoded features, producing the final constructed spectral sequences by the following formulation:

$$\boldsymbol{Y}^{t1} = \omega_2\left(\mathrm{LN}\left(\boldsymbol{X}_3^{t1}\right)\right) \tag{6}$$
$$\boldsymbol{Y}^{t2} = \omega_3\left(\mathrm{LN}\left(\boldsymbol{X}_3^{t2}\right)\right) \tag{7}$$

where ω_2 and ω_3 are trainable matrix, $\boldsymbol{Y}^{t1} \in \mathbb{R}^{c\times 1}$ and $\boldsymbol{Y}^{t2} \in \mathbb{R}^{c\times 1}$ represent the final reconstructed spectral sequences. Thus, the reconstructed spectral sequences in the two temporal are obtained.

B. Cross-Temporal Attention Fusion (CTAF)

By utilizing an advanced interaction learning mechanism, cross-modeling effectively captures temporal correlations across multitemporal images, exhibiting superior performance in HSI-CD [42], [43]. To adequately model long-range dependencies of cross-temporal features in multitemporal HSIs, we similarly incorporate an interactive learning mechanism and propose a novel CTAF module. This module facilitates intertemporal feature interaction by adaptively fusing multitemporal spectral–spatial representations through a self-attention mechanism, thereby capturing discriminative dependencies across heterogeneous temporal domains.

As illustrated in Fig. 4, the proposed CTAF module takes multitemporal sequence features $\boldsymbol{X}_L^{t1} \in \mathbb{R}^{c\times d}$ and $\boldsymbol{X}_L^{t2} \in \mathbb{R}^{c\times d}$ as input. These features are first processed by the cross-temporal attention (CTA) operation to capture intertemporal dependencies and then aggregated via a 1×2 convolutional kernel to generate fused representations. As an example, the CTA computation at time $T2$ is detailed as follows:

$$\boldsymbol{Q}_{t2} = \boldsymbol{X}_L^{t2}W^Q, \boldsymbol{K}_{t1} = \boldsymbol{X}_L^{t1}W^K, \boldsymbol{V}_{t1} = \boldsymbol{X}_L^{t1}W^V$$
$$\boldsymbol{Z}_F = \mathrm{softmax}\left(\frac{\boldsymbol{Q}_{t2}\boldsymbol{K}_{t1}^T}{\sqrt{d_c}}\right)\cdot \boldsymbol{V}_{t1} \tag{8}$$

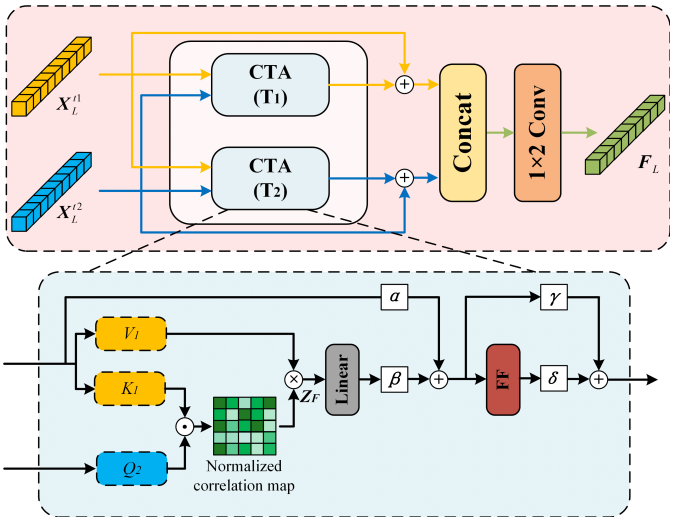


Fig. 4. Internal architecture of CTAF and the detailed computation process of CTA.

$$\boldsymbol{F}'_{t2} = \alpha\cdot\boldsymbol{Z}_F W^O + \beta\cdot\boldsymbol{X}_L^{t1} \tag{9}$$
$$\boldsymbol{F}_{t2} = \gamma\cdot\boldsymbol{F}'_{t2} + \delta\cdot\mathrm{FF}\left(\boldsymbol{F}'_{t2}\right) \tag{10}$$

where $\boldsymbol{Q}_{t2}$ denotes the query at time $t2$, while $\boldsymbol{K}_{t1}$ and $\boldsymbol{V}_{t1}$ represent the key and value from time $t1$, respectively. W^Q, W^K, W^V, and W^O, are four learnable projection matrices. d_c denotes the feature dimension. The parameters α, β, γ, and δ are the corresponding weighting factors. At temporal $t1$, the computation follows a similar procedure as for $t2$, and the output of the CTA module is denoted as $\boldsymbol{F}_{t1} \in \mathbb{R}^{c\times d}$.

The outputs $\boldsymbol{F}_{t1} \in \mathbb{R}^{c\times d}$ and $\boldsymbol{F}_{t2} \in \mathbb{R}^{c\times d}$ from the CTA modules are first concatenated along the third dimension, followed by a convolutional fusion using a 1×2 kernel to generate the final fused feature representation $\boldsymbol{F}_L \in \mathbb{R}^{c\times d}$. The main computational process is as follows:

$$\boldsymbol{F}_L = \mathrm{Conv}(\mathrm{Concat}(\boldsymbol{F}_{t1},\boldsymbol{F}_{t2})). \tag{11}$$

Finally, the fused feature $\boldsymbol{F}_L \in \mathbb{R}^{c\times d}, L \in \{1,2,3\}$ output by the CTAF module is fed into the spectral sequence change analysis module to further refine the CD process.

C. Spectral Sequence Change Analysis

Difference and concatenation are common operations for learning change patterns and generating CD results. To effectively analyze changes in the reconstructed spectral sequences, we employ a difference encoder to extract the difference features between multitemporal sequences. First, the features of the spectral sequences are subtracted, and a randomly generated class token is concatenated along the spectral dimension to form the difference sequence. This can be expressed mathematically as follows:

$$\boldsymbol{D}_0 = \boldsymbol{X}_3^{t1} - \boldsymbol{X}_3^{t2} \tag{12}$$

where $\boldsymbol{D}_0 \in \mathbb{R}^{c\times d}$ is the difference sequence feature.

Then, as shown in Fig. 5, the difference encoder performs deep change analysis on the sequence features to extract difference features. Unlike a decoder, the difference encoder

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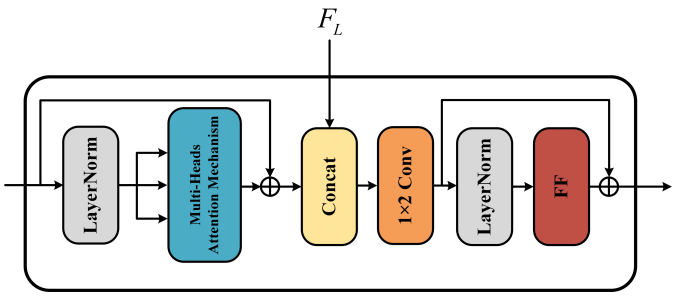


Fig. 5. Internal structure of the difference encoder.

is designed for deep feature encoding. Finally, MLP infers the encoded class token to generate binary CD results, which can be expressed mathematically as follows:

$$\begin{aligned} \boldsymbol{D}_O^{'} &= \mathrm{MHSA}\left(\mathrm{LN}\left(\boldsymbol{D}_{O-1}\right)\right) + \boldsymbol{D}_{O-1} \\ \boldsymbol{D}_O &= \mathrm{FF}\left(\mathrm{LN}\left(\mathrm{Conv}\left(\mathrm{Concat}\left(\boldsymbol{D}_O^{'}, \boldsymbol{F}_L\right)\right)\right)\right) \quad (13) \\ \hat{\boldsymbol{y}} &= \omega_4 \mathrm{LN}\left(\mathrm{Mean}\left(\boldsymbol{D}_3\right)\right) \quad (14) \end{aligned}$$

where ω_4 is a trainable matrix. Mean$(\cdot)$ denotes averaging over the channel dimension, and $\hat{\boldsymbol{y}} \in \mathbb{R}^{1\times 2}$ represents the probabilities of being changed or unchanged. If $\hat{\boldsymbol{y}}(0,1) < \hat{\boldsymbol{y}}(0,0)$, the patch pair is classified as unchanged; otherwise, it is changed.

D. Semi-Supervised HSI-CD Loss Function

The aforementioned sections proposed a semi-supervised HSI-CD framework. To adapt to the SSCD architecture, this section introduces a novel semi-supervised HSI-CD loss function. This loss function consists of a semi-supervised spectral reconstruction loss and a weighted cross-entropy loss. The semi-supervised spectral reconstruction loss uses binary CD results as pseudo-labels to adaptively reduce spectral differences in reconstructed sequences for unchanged areas while preserving those in changed areas. Its primary objective is to introduce a constraint that achieves selective gradient optimization, adaptively amplifying gradients in unchanged areas while attenuating gradients in changed areas, thereby reducing false alarms caused by pseudo-changes. In addition, the weighted cross-entropy loss optimizes training by weighting the cross-entropy based on the distribution of changed and unchanged samples. Its primary objective is to mitigate the long-tail problem by leveraging the sample distribution across different datasets.

Specifically, the semi-supervised spectral reconstruction loss can weigh the differences between the two spectral sequences based on the binary CD results output. This ensures that spectral differences in unchanged areas are reduced while preserving those in changed areas. The semi-supervised spectral reconstruction loss is defined as follows:

$$\begin{aligned} \boldsymbol{Loss}_{t1} &= \mathrm{MSE}\left(\mathrm{Mean}\left(\boldsymbol{X}^{t1}\right)\cdot \hat{\boldsymbol{y}}\left(0,0\right), \boldsymbol{Y}^{t2}\cdot \hat{\boldsymbol{y}}\left(0,0\right)\right) \\ \boldsymbol{Loss}_{t2} &= \mathrm{MSE}\left(\mathrm{Mean}\left(\boldsymbol{X}^{t2}\right)\cdot \hat{\boldsymbol{y}}\left(0,0\right), \boldsymbol{Y}^{t1}\cdot \hat{\boldsymbol{y}}\left(0,0\right)\right) \\ \boldsymbol{Loss}_R &= \boldsymbol{Loss}_{t1} + \boldsymbol{Loss}_{t2} \quad (15) \end{aligned}$$

where $\boldsymbol{Loss}_R$ is the semi-supervised spectral reconstruction loss, MSE represents the mse loss. Mean denotes the mean calculation over the patch dimension. $\boldsymbol{Y}^{t1}$ and $\boldsymbol{Y}^{t2}$ correspond to the reconstructed spectral sequences in temporal one and temporal two, respectively. $\boldsymbol{X}^{t1} \in \mathbb{R}^{c\times p\times p}$ and $\boldsymbol{X}^{t2} \in \mathbb{R}^{c\times p\times p}$ represent the spectral sequences of the HSI patches of temporal one and temporal two, respectively. Lastly, $\boldsymbol{y}(0,0)$ indicates the probability inferred by the MLP that the corresponding pixel belongs to the unchanged area.

From (15), it is clear that when the model predicts a pixel as unchanged, $\boldsymbol{y}(0,0)$ approaches 1, and $\boldsymbol{Loss}_R$includes the mse between the corresponding spectral sequences. In this case, the loss function provides effective gradients to guide the model in minimizing spectral difference within unchanged areas. Conversely, when the network predicts a pixel as changed, resulting in the $\boldsymbol{y}(0,0)$ also approaching 0, resulting in the $\boldsymbol{Loss}_R$ also approaching 0, effectively excluding the mse for changed pixels from the loss.

Furthermore, in CD datasets, the distribution of changed samples and unchanged samples is often imbalanced. In the same geographic area, the area of unchanged regions is typically larger compared with changed regions, which usually occupy only a small portion of the total affected area. To balance the influence of changed and unchanged samples on training, we propose a weighted cross-entropy loss. This loss adjusts the cross-entropy weights based on the distribution of changed and unchanged samples, thereby optimizing training. It is defined as follows:

$$\begin{aligned} w_c &= \frac{1-\lambda}{1-\lambda^{n_c}}/2, w_{uc} = \frac{1-\lambda}{1-\lambda^{n_uc}}/2 \\ \mathrm{Loss}_w &= -\frac{1}{N}\sum_{i=1}^{N} w_c \cdot \boldsymbol{y}_i\left(0,1\right)\cdot \log\left(\hat{\boldsymbol{y}}\left(0,1\right)\right) \\ &\quad + w_{\mathrm{uc}} \cdot \boldsymbol{y}_i\left(0,0\right)\cdot \log\left(\hat{\boldsymbol{y}}\left(0,0\right)\right) \quad (16) \end{aligned}$$

where w_c and w_{uc} present the weights for change samples and nonchange samples, respectively. n_c and $n_$uc denote the total number of change pixels and unchanged pixels in the dataset, respectively. The parameter λ is manually set to 0.99999 in this section. $\boldsymbol{y}(0,1))$ and $\boldsymbol{y}(0,0))$ correspond to the labels of the respective pixels in the ground truth image. $\hat{\boldsymbol{y}}(0,1)$ and $\hat{\boldsymbol{y}}(0,0)$ represent the probabilities inferred by the model for a pixel being changed or unchanged, respectively. Finally, N is the total number of pixels.

In summary, the semi-supervised HSI-CD loss function proposed in this section consists of the semi-supervised spectral reconstruction loss $\boldsymbol{Loss}_R$ and the weighted cross-entropy loss $\boldsymbol{Loss}_w$. This can be mathematically expressed as follows:

$$\boldsymbol{Loss}_{\mathrm{semi}} = \boldsymbol{Loss}_R + \boldsymbol{Loss}_w. \quad (17)$$

For the implementation mechanism of CSR-Net in the program, a pseudocode is provided in Algorithm 1.

IV. Experiment

To validate the performance of CSR-Net with limited labeled samples, we conducted experiments using only 100 labeled samples on four publicly available HSI-CD datasets. First, four well-known HSI datasets are described in detail. Second, we provide a description of the experimental setting, including the evaluation criteria, comparative experiment, and

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Algorithm 1 CSR-Net for HSI-CD

Input: $\boldsymbol{X}^{t1} \in \mathbb{R}^{c\times p\times p}, \boldsymbol{X}^{t2} \in \mathbb{R}^{c\times p\times p}$ from multitemporal HSI.
Output: Changed map $\hat{\boldsymbol{y}} \in \mathbb{R}^{1\times 2}$
#Optimization with Deep Network-in-Training
while $j <$ epoch **do**
  #Feature embedding and position embedding.
  $\boldsymbol{X}_0^{t1} = \omega_1 \boldsymbol{X}^{t1} + PE, \boldsymbol{X}_0^{t2} = \omega_1' \boldsymbol{X}^{t2} + PE$
    #Feature encoding in spectral reconstruction sub-network and CTAF module
    for $L \in \{1, 2, 3\}$ **do**
    #Feature encoding in spectral encoder.
    $\boldsymbol{X}_L^{t1} = \mathrm{FF}(\mathrm{MHSA}(\mathrm{LN}(\boldsymbol{X}_{L-1}^{t1})) + \boldsymbol{X}_{L-1}^{t1})$
    $\boldsymbol{X}_L^{t2} = \mathrm{FF}(\mathrm{MHSA}(\mathrm{LN}(\boldsymbol{X}_{L-1}^{t2})) + \boldsymbol{X}_{L-1}^{t2})$
    #CTAF
    $\boldsymbol{Q}_{t2} = \boldsymbol{X}_L^{t2} W^Q, \boldsymbol{K}_{t1} = \boldsymbol{X}_L^{t1} W^K, \boldsymbol{V}_{t1} = \boldsymbol{X}_L^{t1} W^V$
    $\boldsymbol{Z}_F = \mathrm{softmax}\left(\frac{\boldsymbol{Q}_{t2}\boldsymbol{K}_{t1}^T}{\sqrt{d_c}}\right) \cdot \boldsymbol{V}_{t1}$
    $\boldsymbol{F}'_{t2} = \alpha \cdot \boldsymbol{Z}_F W^O + \beta \cdot \boldsymbol{X}_L^{t1}$
    $\boldsymbol{F}_{t2} = \gamma \cdot \boldsymbol{F}'_{t2} + \delta \cdot \mathrm{FF}(\boldsymbol{F}'_{t2})$
    $\boldsymbol{F}_L = \mathrm{Conv}(\mathrm{Concat}(\boldsymbol{F}_{t1}, \boldsymbol{F}_{t2}))$
  end for
  #Spectral sequence reconstruction
  $\boldsymbol{Y}^{t1} = \omega_2(\mathrm{LN}(\boldsymbol{X}_3^{t1})), \boldsymbol{Y}^{t2} = \omega_3(\mathrm{LN}(\boldsymbol{X}_3^{t2}))$
  #Difference map
  $\boldsymbol{D}_0 = \boldsymbol{X}_3^{t1} - \boldsymbol{X}_3^{t2}$
  for $O \in \{1, 2, 3\}$ **do**
    #Feature encoding in Difference encoder.
    $\boldsymbol{D}'_O = \mathrm{MHSA}(\mathrm{LN}(\boldsymbol{D}_{O-1})) + \boldsymbol{D}_{O-1}$
    $\boldsymbol{D}_O = \mathrm{FF}(\mathrm{LN}(\mathrm{Conv}(\mathrm{Concat}(\boldsymbol{D}'_O, \boldsymbol{F}_L))))$
  end for
  #Feature encoding in Difference encoder
  $\hat{\boldsymbol{y}} = \omega_4 \mathrm{LN}(\mathrm{Mean}(\boldsymbol{D}_3))$
  #Calculate Loss
  $\boldsymbol{Loss}_{t1} = l_{MSE}(\mathrm{Mean}(\boldsymbol{X}^{t1}) \cdot \hat{\boldsymbol{y}}(0,0), \boldsymbol{Y}^{t2} \cdot \hat{\boldsymbol{y}}(0,0))$
  $\boldsymbol{Loss}_{t2} = l_{MSE}(\mathrm{Mean}(\boldsymbol{X}^{t2}) \cdot \hat{\boldsymbol{y}}(0,0), \boldsymbol{Y}^{t1} \cdot \hat{\boldsymbol{y}}(0,0))$
  $\boldsymbol{Loss}_R = \boldsymbol{Loss}_{t1} + \boldsymbol{Loss}_{t2}$
  $w_c = \frac{1-\lambda}{1-\lambda^{n-c}}/2, w_{uc} = \frac{1-\lambda}{1-\lambda^{n-uc}}/2$
  $Loss_{\mathrm{w}} = -\frac{1}{N}\sum_{i=1}^{N} w_c \cdot \boldsymbol{y}_i(0,1) \cdot \log(\hat{\boldsymbol{y}}(0,1)) + w_{uc} \cdot \boldsymbol{y}_i(0,0) \cdot \log(\hat{\boldsymbol{y}}(0,0))$
  $\boldsymbol{Loss}_{semi} = \boldsymbol{Loss}_R + \boldsymbol{Loss}_w$
end while
#Testing Stage for the CSR-Net.
Import Multitemporal HSI patches into CSR-Net, and obtain the change map $\hat{\boldsymbol{y}}$, if $\hat{\boldsymbol{y}}(0,1) < \hat{\boldsymbol{y}}(0,0)$, the CD result of the patch pair is unchanged; otherwise, it is changed.

hyperparameters of CSR-Net. Third, we compare the performance between our proposed method and other state-of-the-art methods for SSCD. Finally, we conduct ablation studies to isolate component contributions.

A. Experimental HSI-CD Datasets

The first dataset used in the experiment is the Farmland dataset, which was acquired by the Earth Observing-1 (EO-1) hyperion hyperspectral sensor on May 3, 2006, and April 23, 2007. The pseudo-color images are shown in Fig. 6. The

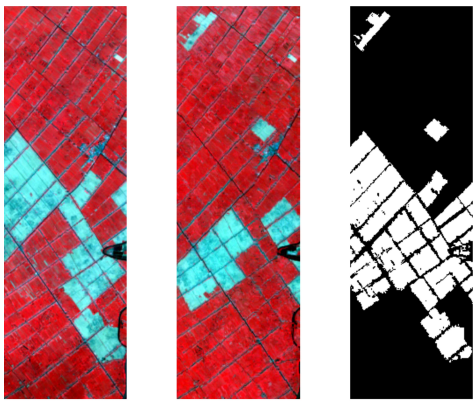


Fig. 6. Farmland dataset, where (a) temporal-one image acquired on May 3, 2006, and (b) temporal-two image acquired on April 23, 2007, and (c) ground truth.



Fig. 7. Santa Barbara dataset, where (a) temporal-one image acquired in 2013, (b) temporal-two image acquired in 2014, and (c) ground truth.

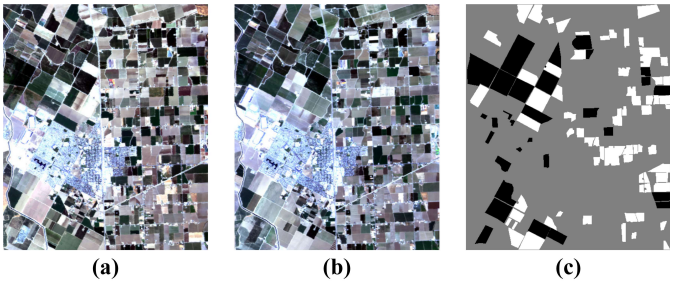


Fig. 8. Bay Area dataset, where (a) temporal-one image acquired in 2013, (b) temporal-two image acquired in 2014, and (c) ground truth.

size of the multitemporal HSIs is 420×140 pixels with 30-m spatial resolution. After noisy band removal, they contain 155 spectral bands covering the spectral range of 0.4–2.5 μm.

The second dataset, the Santa Barbara dataset, was obtained in 2013 and 2014 by using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The HSIs have a size of 984×740 pixels with a spatial resolution of 20-m. After noisy band removal, the two HSIs are both characterized by 224 bands covering the spectral range of 0.4–2.5 μm. The pseudo-color images are shown in Fig. 7. The ground-truth image is composed of 80 418 unchanged pixels, 52 134 changed pixels, and 595 608 undetermined pixels.

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TABLE I
NUMBER OF PIXEL PAIRS IN TRAINING AND TESTING SETS FOR THE FOUR DATASETS

Dataset	Labeled Training Samples			Unlabeled Training Samples			Testing Samples		
	Unchanged	Changed	Total	Unchanged	Changed	Total	Unchanged	Changed	Total
Farmland	100	100	200	1000	1000	2000	17177	43623	60800
Santa Barbara	100	100	200	1000	1000	2000	79318	51034	130352
Bay Area	100	100	200	1000	1000	2000	33111	38170	71281
Hermiston	100	100	200	1000	1000	2000	66914	8886	75800

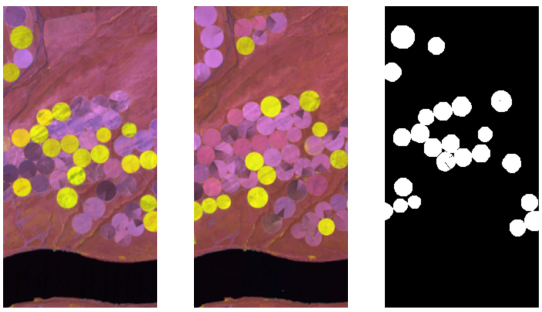


Fig. 9. Hermiston dataset, where (a) temporal-one image acquired in 2004, (b) temporal-two image acquired in 2007, and (c) ground truth.

The third Bay Area dataset was obtained in 2013 and 2014 by the AVIRIS sensor. The pseudo-color images are shown in Fig. 8. The size of the Bay Area dataset is 600 × 500 pixels with 20-m spatial resolution, and the two HSIs both contain 224 bands in the 0.4-2.5 μm spectral region. There are 38 425 changed pixels, 34 211 unchanged pixels, and 227 364 undetermined pixels in the ground-truth image.

The fourth Hermiston dataset was acquired on May 1, 2004, and May 8, 2007, using the HYPERION sensor over the Hermiston City area in Oregon. The dataset has spatial dimensions of 390 × 200 pixels and includes 242 spectral bands. The pseudo-color images are shown in Fig. 9.

The CD ground-truth of the above four datasets is shown in Figs. 6(c), 7(c), 8(c), and 9(c), respectively. In these ground-truth images, white, black, and gray pixels represent changed, unchanged, and undetermined regions, respectively.

B. Experimental Setup

We evaluate the CD performance of each method quantitatively in terms of two commonly used indices, including overall accuracy (OA) and Kappa coefficient (Kappa), which are calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{18}$$

$$\mathrm{P_C} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2} \tag{19}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{P_C}}{1 - P_C} \tag{20}$$

where true positive (TP) is the number of correctly detected changes in pixels, true negative (TN) is the number of correctly detected unchanged pixels, false positive (FP) represents the number of pixels changed by error detection, and false negative (FN) is the number of unchanged pixels in error detection.

To verify the superiority of CSR-Net, we choose eight representative methods for comparison: CVA, SVM, recurrent 3D fully convolutional network (Re3FCN) [44], spectral-spatial-temporal transformer (SST-Former) [45], spectral token guidance transformer (STCD-Former) [28], domain alignment transformer (DAFormer) [46], diff-feature contrast enhancement network (DCE-Net) [47], and adaptive information weighting and synchronized enhancement network (AIWSEN) [48]. We use the same training samples to train the above networks for fairness. As shown in Table I, 100 labeled samples are selected from both the changed and unchanged categories, while 1000 unlabeled samples are taken from each category as well. The remaining samples are reserved for validation and testing. The other details in the experiment are as follows.

CVA was used to obtain the change vector and automatically classified change and nonchange by K-means.

When applying SVM for HSI-CD, we first used PCA to extract the principal components of the difference map, and used the SVM to detect changes, which is achieved by the function sklearn.svm in Python.

Re3FCN consists of two convolutional layers and recurrent layers with an LSTM module to extract spatial–temporal information for HSI-CD. The detailed setting can be found in their article.

Within the Transformer-based methods, SST-Former extracts spatial–spectral–temporal sequential information to produce CD results. STCD-Former employs an ST-Former module to generate spectral tokens, followed by spatial interaction operations for CD. DA-Former utilizes a Domain Alignment Transformer to align multitemporal images for executing CD. The patch size was configured as 7, with all other parameters consistent with those reported in the original publications.

DCE-Net employs a differential fusion attention mechanism to extract temporal features from the HSI patches. AIWSEN employs the information entropy to capture change features specific to the CD task and enhances bitemporal interaction features. In our replication study, these identical semi-supervised architectures and loss functions were implemented, with all other experimental configurations maintained as specified in the original publication.

Our proposed CSR-Net was implemented on PyTorch 1.11.1 with an Intel Xeon Silver 4210R CPU and an NVIDIA GeForce RTX 3090 24 G GPU. The learning rate of the Adam optimizer was set to 0.0005. We used ExponentialLR to update the learning rate by multiplying 0.997 after each epoch. We set the epochs to 500 and used Sigmoid as the activation function to output the binary CD probabilities. Furthermore,

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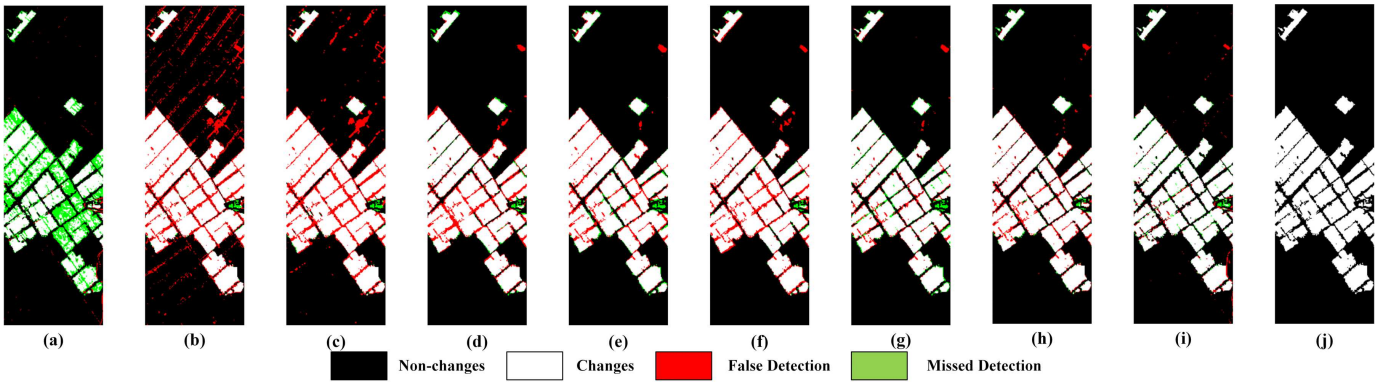


Fig. 10. CD results on the Farmland dataset. (a) CVA. (b) SVM. (c) Re-3FCN. (d) SST-Former. (e) STCD-Former. (f) DA-Former. (g) DCE-Net. (h) AIWSEN. (i) CSR-Net. (j) ground truth. The white is the changed area. The black is the unchanged area.

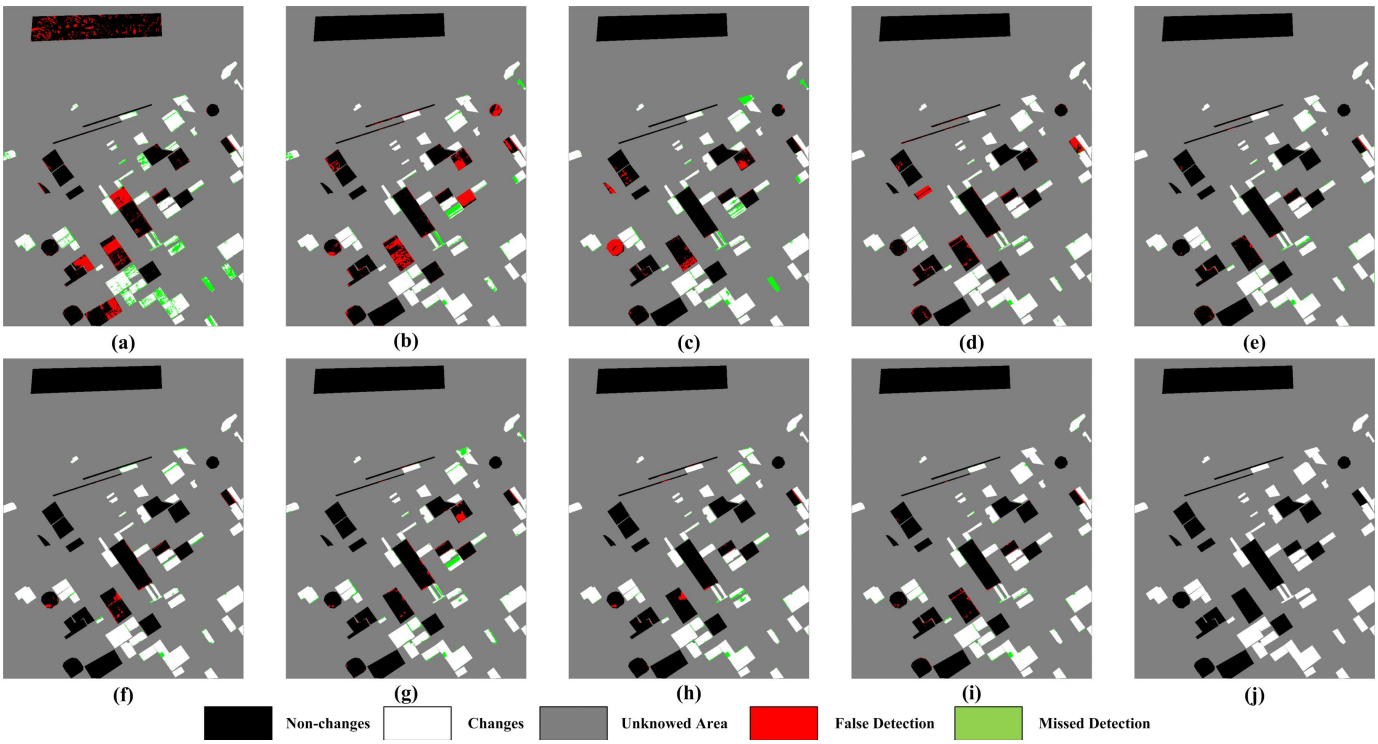


Fig. 11. CD results on Santa Barbara dataset. (a) CVA. (b) SVM. (c) Re-3FCN. (d) SST-Former. (e) STCD-Former. (f) DA-Former. (g) DCE-Net. (h) AIWSEN. (i) CSR-Net. (j) Ground truth. The white area is the change. The black area is unchanged. And the gray is the unknown area.

following experimental parameter optimization, the patch size is configured as 1 for the Farmland dataset, 5 for both the Santa Barbara and Bay Area datasets, and 3 for the Hermiston dataset. The number of reconstruction encoders and change analysis encoders is set to 3 and 3, respectively. The batch size is set to 64.

C. Comparison With Other Methods

To evaluate the performance of our proposed method, the other eight representative methods are compared with CSR-Net both qualitatively (via visualization) and quantitatively.

All these methods were tested five times on different datasets, and the qualitative results for the four datasets are shown in Table II. Figs. 10–13 list the visualization results for the four datasets, respectively. Furthermore, all results for CSR-Net are reported as the mean ± standard deviation, highlighting its enhanced stability.

1) Analysis for the Farmland Dataset: As shown in Table II, the traditional CD algorithm, CVA, exhibits the lowest performance metrics. Moreover, Fig. 10(a) shows that the CVA results contain numerous undetected areas (marked in green). In addition, the classical machine learning algorithm SVM also performs poorly in OA and Kappa when trained with minimal samples. As a representative CNN-based CD method, Re3FCN achieves better performance than CVA. However, Fig. 10(c) shows that the Re3FCN results include numerous false detections (marked in red). The Transformer-based methods SST-Former and STCD-Former achieve superior CD performance but also show false detections in Fig. 10(d) and (e). In addition, the semi-supervised HSI-CD algorithms DA-Former and DCE-Net achieve impressive performance. Lastly, AIWSEN achieves a Kappa coefficient of 91.15% with very limited samples. However, CSR-Net outperforms all other methods, achieving

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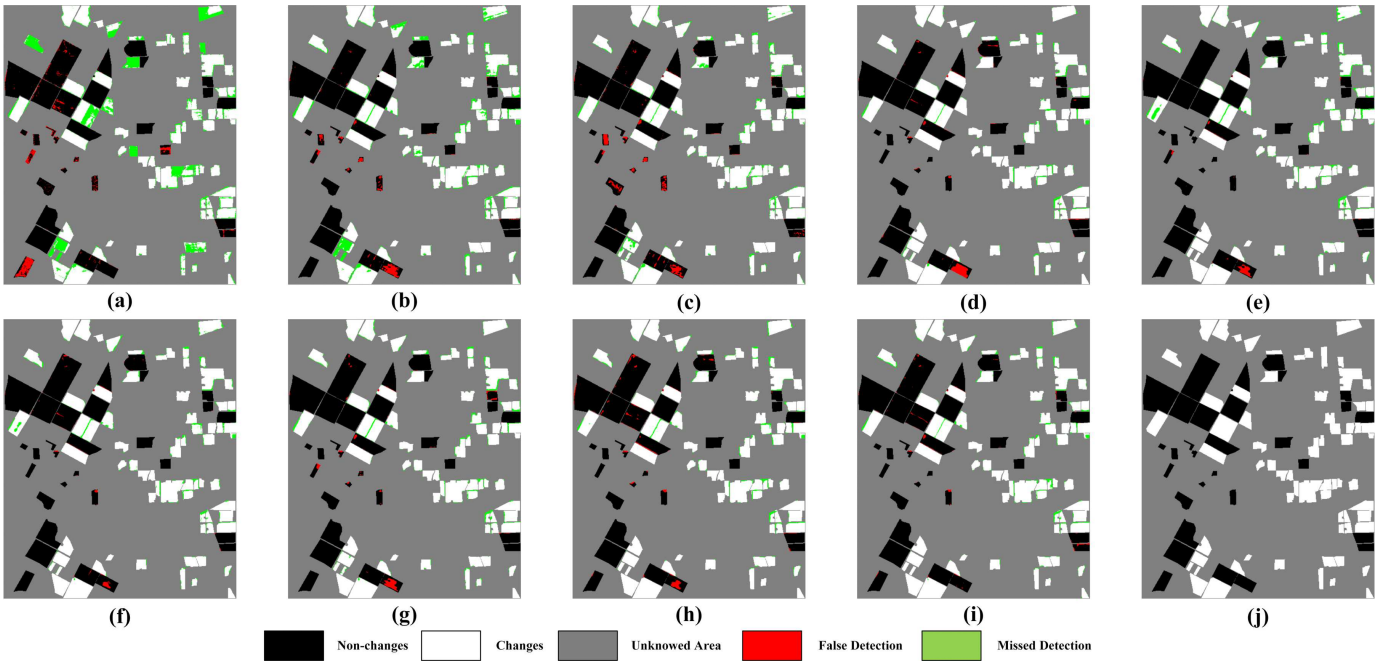


Fig. 12. CD results on Bay Area dataset. (a) CVA. (b) SVM. (c) Re-3FCN. (d) SST-Former. (e) STCD-Former. (f) DA-Former. (g) DCE-Net. (h) AIWSEN. (i) CSR-Net. (j) Ground truth. The white area is the change. The black area is unchanged. And the gray is the unknown area.

TABLE II
ACCURACY COMPARISON OF EXCELLENT HSI-CD METHODS ON FOUR DATASETS

Dataset	Metrics	CVA	SVM	Supervised-based HSI-CD				Semi-Supervised-based HSI-CD		
				Re3FCN	SST-Former	STCD-Former	AIWSEN	DCE-NET	DA-Former	OURS
Farmland	OA (%)	88.82	90.00	91.91	94.47	94.94	96.26	96.02	94.48	**97.67**±0.77
	Kappa (%)	69.78	77.86	81.74	87.10	88.05	91.15	90.41	87.21	**94.34**±0.32
Santa Barbara	OA (%)	83.21	90.23	92.06	95.23	97.34	97.06	95.47	97.46	**97.64**±0.41
	Kappa (%)	65.09	79.85	83.31	90.06	94.41	93.79	90.44	94.67	**95.04**±0.19
Bay Area	OA (%)	84.00	90.89	93.61	95.91	95.54	96.09	95.79	96.35	**96.81**±0.79
	Kappa (%)	68.32	81.85	87.20	91.80	91.08	92.16	91.56	92.70	**93.61**±0.31
Hermiston	OA (%)	93.05	95.40	96.03	97.03	98.07	97.94	96.09	98.18	**98.81**±0.39
	Kappa (%)	74.19	81.63	84.16	87.78	91.70	91.27	84.19	92.23	**94.79**±0.31

the highest CD accuracy. Its Kappa coefficient surpasses that of AIWSEN by nearly 3.3%.

2) Analysis for the Santa Barbara Dataset: Table II also presents comparative experiment results on the Santa Barbara dataset. CVA, as a mainstream method in algebraic CD, also shows inferior performance compared with machine learning and DL methods. In addition, SVM achieves better CD performance than CVA but still falls short of CNN-based methods such as Re3FCN. Among Transformer-based methods, SST-Former achieves a Kappa coefficient of 90.06%. STCD-Former and DA-Former both achieve Kappa coefficients above 94%. However, DCE-Net achieves a Kappa coefficient approximately 4% lower. AIWSEN achieves an OA of 97.06% and a Kappa of 93.79%, which are lower than those of the Transformer-based methods. Finally, CSR-Net achieves the best overall performance, surpassing the second-place method by 0.37% in Kappa. Furthermore, Fig. 11 shows that CSR-Net exhibits fewer missed detections and FPs.

3) Analysis for the Bay Area Dataset: In Table II, CVA and SVM also achieve the lowest accuracy. Re3FCN attains a Kappa of 87.20%, which is higher than SVM. Among Transformer-based methods, SST-Former achieves a Kappa of 91.80%, approximately 0.8% higher than STCD-Former. In addition, DCE-Net outperforms STCD-Former by nearly 0.2% in Kappa. AIWSEN achieves a Kappa of 92.16%. Among semi-supervised CD methods, the proposed CSR-Net exceeds DA-Former by nearly 0.5% in OA and 0.9% in Kappa. In the visualization results (see Fig. 12), CSR-Net exhibits the fewest false detections (marked in red), whereas other methods show false detections in the lower image regions.

4) Analysis for the Hermiston Dataset: As shown in Table II, the traditional CD algorithm CVA yields the lowest performance metrics. Fig. 13(a) shows that the CVA results contain numerous undetected areas. In addition, SVM achieves a high OA, but Fig. 13(b) shows significant false detections over large areas. As a representative CNN-based CD method, Re3FCN exhibits superior performance compared with CVA. However, Fig. 13(c) shows that the Re3FCN results contain considerable FPs. Transformer-based methods such as SST-Former and STCD-Former demonstrate enhanced performance

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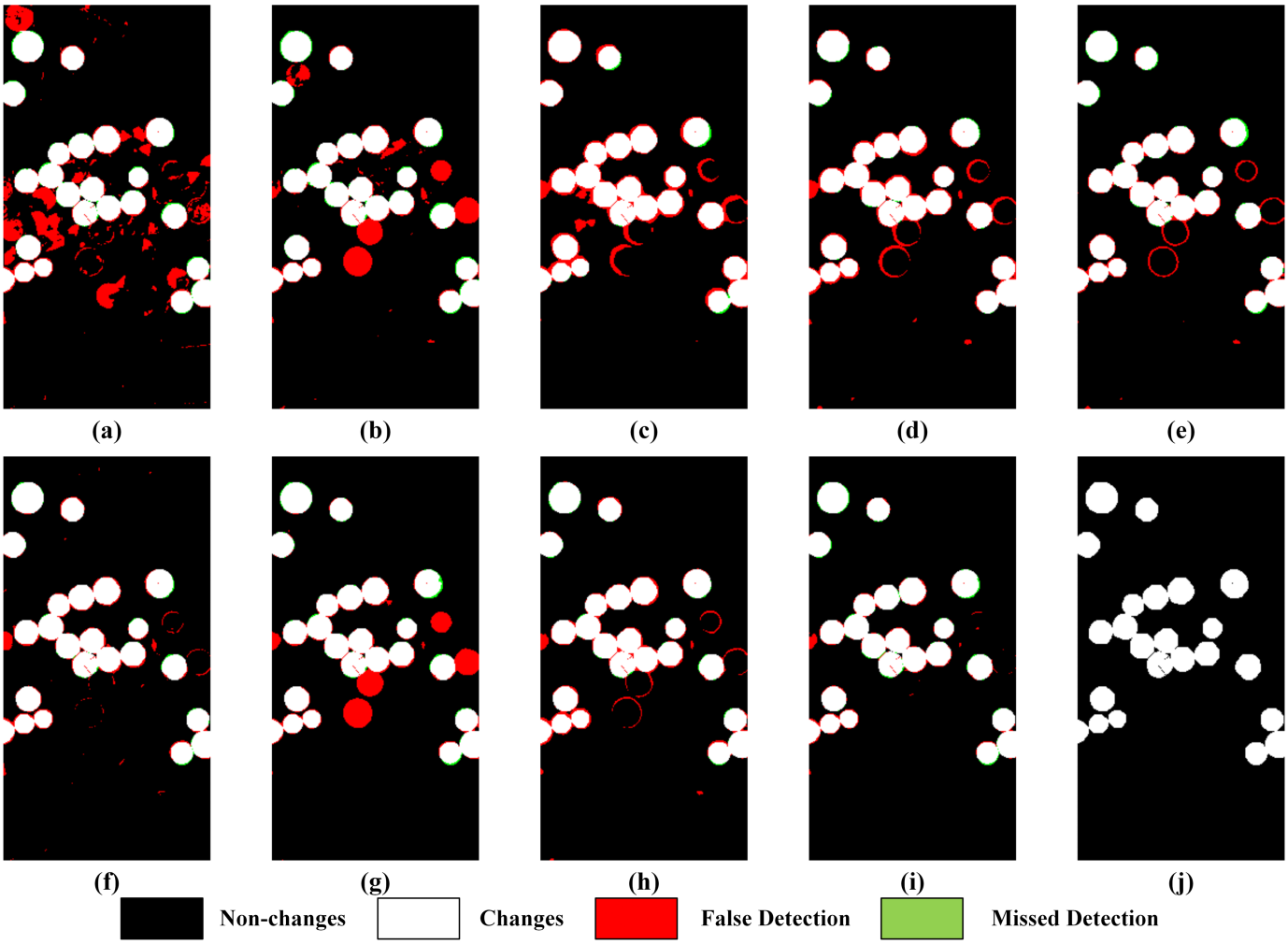


Fig. 13. CD results on Hermiston dataset. (a) CVA. (b) SVM. (c) Re-3FCN. (d) SST-Former. (e) STCD-Former. (f) DA-Former. (g) DCE-Net. (h) AIWSEN. (i) CSR-Net. (j) Ground truth. The white is the changed area. The black is the unchanged area.

TABLE III
COMPARISON OF EFFICIENCY ON THE FARMLAND DATASET

Model	Params (M)	FLOP (G)
SST-Former	2.5	14.47
STCD-Former	2.73	15.3
DA-Former	0.4	0.04
AIWSEN	12.37	124.82
OURS	0.4	2.4

but also introduce false detections, as shown in Fig. 13(d) and (e).

The semi-supervised method DA-Former achieves commendable performance. However, Fig. 13(g) shows that DCE-Net exhibits significant false detections over large areas. AIWSEN achieves a Kappa of 91.27% with very limited samples. Nevertheless, CSR-Net outperforms all other methods, achieving the highest accuracy. Its Kappa exceeds that of AIWSEN by nearly 3.5%. Fig. 13(h) and (i) shows that CSR-Net results in fewer FPs and missed detections, confirming its efficacy in fine-grained CD.

5) Comparison of Efficiency: As summarized in Table III, the proposed CSR-Net maintains a compact structure with only 0.40M parameters and 2.40 G floating point operations (FLOPs). While DA-Former exhibits lower computational overhead, it is consistently outperformed by our method across all evaluation metrics, as evidenced by the results in Table II. In summary, CSR-Net achieves a compelling balance between computational efficiency and detection accuracy, thereby demonstrating its strong practical potential for the HSI-CD task.

D. Ablation Analysis

Our proposed CSR-Net mainly consists of five parts: A—cross-temporal spectral sequence reconstruction; B— spectral sequence change analysis; C—CTAF; D—semi-supervised spectral reconstruction loss; and E—weighted cross-entropy loss. We conducted an ablation study to validate the effectiveness of each component.

For component A, we use dual-branch spectral reconstruction encoders to reconstruct the spectral sequences and an MLP head to predict changes from the reconstructed sequences. For component B, we directly subtract the mul-

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TABLE IV
ABLATION RESULTS ON FARMLAND DATASET

Model	A	B	C	D	E	Kappa (%)
CSR-Net		✓	✓	✓	✓	91.61
	✓		✓	✓	✓	90.64
	✓	✓		✓	✓	93.51
	✓	✓	✓		✓	93.06
	✓	✓	✓	✓		93.01
	✓	✓	✓	✓	✓	94.34

titemporal HSIs to obtain a difference image, which is then encoded by a spectral difference encoder. Finally, an MLP predicts the difference features for CD.

Table IV shows that using only component A or B yields unsatisfactory Kappa values, indicating that spectral sequence reconstruction or change analysis alone is insufficient. In addition, the CTAF module (C) enhances CD performance, improving the Kappa by nearly 1%. To validate the effectiveness of the loss functions (D and E), we independently integrated each into CSR-Net for training and evaluation. As shown in Table IV, using only the weighted cross-entropy loss (E) for supervised training significantly enhances model performance. Incorporating the semi-supervised spectral reconstruction loss (D) further improves performance. Removing the weight optimization from the weighted cross-entropy loss reduces the Kappa by 1.33%. Moreover, to validate the robustness of the weighted cross-entropy loss, experiments were conducted on the Farmland dataset with artificially balanced changed/unchanged samples (including 182 changed samples and 442 unchanged samples). The model achieves consistent performance (OA: 98.34%, Kappa: 95.94%), confirming the loss's effectiveness across different class distributions.

To validate the superiority of the proposed semi-supervised loss, we incorporated it into the training of DA-Former. Experimental results show that the Kappa value increases significantly from 87.21% to 89.62%. These findings demonstrate the effectiveness and generalizability of the proposed semi-supervised loss.

In addition, to validate the impact of unidirectional loss versus bidirectional loss on model accuracy, we removed one loss term from (15) and used only the remaining term as a constraint for model training. Experimental results demonstrate that, on the Farmland dataset, the model accuracy with unidirectional loss (Kappa) is 1.36% lower than that with bidirectional loss. The bidirectional loss provides additional gradients, enabling the model to better learn the spectral–temporal relationships.

E. Effect of Hyperparameters

In our proposed CSR-Net, some hyperparameters affect the training process and CD performance. Here, we examine the number of labeled and unlabeled training samples, patch size, and batch size. Thus, we investigated the influence of these hyperparameters in this section. In this process, we fixed the other parameters when analyzing the impact of one certain parameter.

TABLE V
QUANTITATIVE COMPARISON OF CROSS-DATASET (FARMLAND TO BAY AREA) VALIDATION

Metrics	AIWSEN	CSR-Net
OA (%)	52.8	63.61
Kappa (%)	9.81	25.69

1) Number of Labeled Training Samples: The set of numbers 10, 50, 100, 200, and 300 was selected in this experiment. As shown in Fig. 14(a), the more labeled training samples we use, the higher accuracy we achieve.

2) Number of Unlabeled Training Samples: We experimented with the number of unlabeled training samples with 5 numbers {100, 300, 500, 1000, 2000}. As shown in Fig. 14(b), we found that the number of unlabeled training samples also impacts model training. With an increased quantity of unlabeled samples provided, the model can approximately gain a 2% improvement in the Kappa coefficient.

3) Patch Size: We examined five patch sizes {1×1, 3×3, 5×5, 7×7, 9×9} to analyze the influence of the patch size. As illustrated in Fig. 14(c), a patch size of 7×7 yielded nearly optimal results on the Santa Barbara and Bay Area datasets. However, for the Farmland dataset, the 1×1 patch size outperformed the others. Consequently, we selected 5×5 as the input patch size for Santa Barbara and Bay Area, and 1×1 for the Farmland dataset. Moreover, while larger patch sizes theoretically enhance contextual information, they may lead to spectral over-averaging in fine-grained detection scenarios. In the case of the Farmland dataset, smaller patches (1×1) excel at capturing localized agricultural changes while suppressing pseudo-changes, achieving a 9.11% higher Kappa compared with 9×9 patches. This finding is consistent with pseudo-labeling strategies [49], where more granular initial labels contribute to improved iterative refinement.

4) Batch Size: To evaluate the influence of the batch size on CSR-Net, a set of batch sizes {8, 16, 32, 64, 128} was considered. As shown in Fig. 14(d), the Kappa values for different batch sizes are similar; we set the batch size to 64.

F. Cross-Dataset Evaluation

To further validate the generalization capability of CSR-Net and its potential for real-world application, this section incorporates a cross-dataset validation experiment. The model is first trained on the Farmland dataset and then tested on the Bay Area dataset. The cross-dataset validation results are presented in Table V. Although CSR-Net outperforms other methods, it achieves only 63.61% OA. The results demonstrate that while CSR-Net performs well in dataset-specific training/testing splits, its performance degrades significantly in cross-dataset scenarios.

The Farmland and Bay Area datasets were acquired by different sensors with incompatible band counts, spatial resolutions, and land cover compositions. This fundamental sensor discrepancy invalidates direct cross-dataset comparison or adaptation. Even datasets with identical sensor specifications—such as Bay Area and Santa Barbara, both

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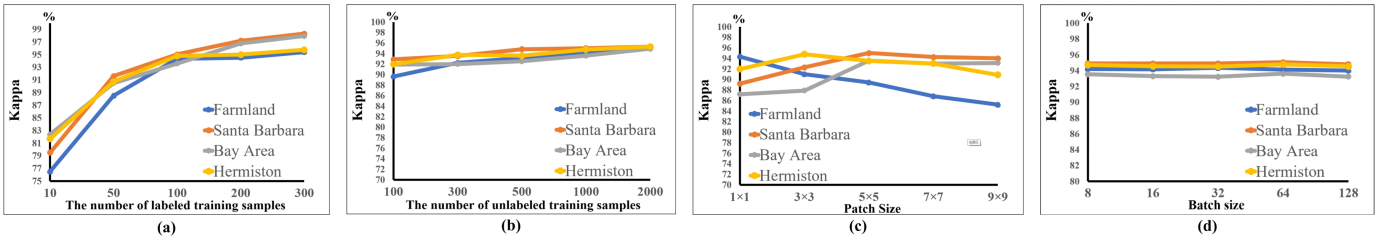


Fig. 14. Accuracy of the proposed CSR-Net with different parameters on four datasets. (a) Number of labeled training samples. (b) Number of unlabeled training samples. (c) Patch size. (d) Batch size.

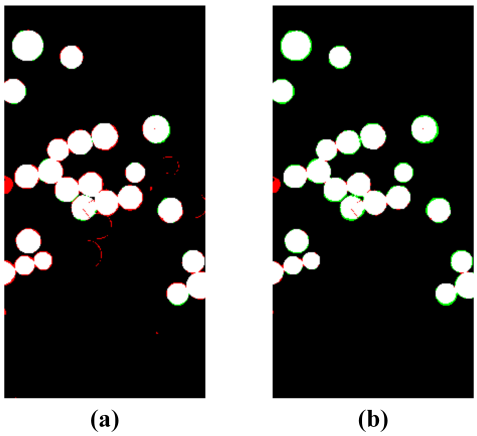


Fig. 15. Comparison of CD results on the Hermiston dataset by CSR-Net before and after morphological edge refinement. (a) Initial change map. (b) Refined change map.

from AVIRIS—exhibit substantial differences in land cover, further complicating generalization.

Based on the above discussion, similar to the vast majority of existing HSI-CD methods, CSR-Net does not perform well in cross-dataset settings, which also limits its practicality in real-world scenarios. Therefore, developing full domain generalization approaches represents an important direction for future work.

G. Post-Processing for Edge Refinement

Although CSR-Net demonstrates strong performance across multiple datasets, minor inaccuracies persist along certain land-cover boundary regions. Edge detection errors, as observed in Fig. 13(i), primarily stem from spectral mixing effects, where mixed pixels obscure legitimate changes; this represents a perennial challenge in HSI processing. To mitigate such spurious responses in boundary regions, lightweight post-processing strategies—such as morphological edge refinement, guided filtering, or spectral–spatial consistency constraints—can effectively reduce false detections. In this work, we introduce a morphological post-processing step involving closing and opening operations with a 3×3 kernel to enhance the practical utility of CSR-Net for applications requiring precise object delineation. As illustrated in Fig. 15, which compares results before and after edge refinement on the Hermiston dataset, the refined output in Fig. 15(b) exhibits reduced false detections along boundaries.

Based on the above discussion, employing edge refinement post-processing proves highly practical for HSI-CD. Nevertheless, like most HSI-CD methods, CSR-Net remains affected by spectral mixing effects, leading to certain false detections in boundary regions with mixed pixels. Therefore, future research could incorporate nonlinear unmixing techniques to separate mixed pixels and improve boundary delineation.

V. Discussion

Semi-supervised HSI-CD is a crucial and challenging task that involves extracting supervisory information from unlabeled samples to optimize model training, thereby enhancing CD performance. Specifically, the spectral differences between unchanged areas in multitemporal HSI can lead to false detections. To address this issue, a distinct reconstruction-guided approach for semi-supervised HSI-CD is proposed. Experiments are carried out on four commonly used HSI-CD datasets. The results show that the proposed framework achieves state-of-the-art performance. However, some key issues need to be discussed for further application of CSR-Net.

1) Most existing HSI-CD methods preprocess the input HSI by dividing it into multiple patches for model input, with this preprocessing module executed on the CPU. Our method adopts a similar preprocessing approach. This results in increased memory usage and longer processing times for larger HSIs. We plan to deploy the preprocessing module on a GPU in the future, leveraging parallel computing to accelerate processing speeds.
2) Although the semi-supervised CD model proposed in this article can learn effective supervision information from a small number of labeled samples and a large number of unlabeled samples, it still requires a small number of labeled samples for training. However, in many real-world scenarios, such as after natural disasters, it is challenging to obtain sufficient labeled samples in a timely manner for training. Therefore, future work will integrate unsupervised and self-supervised learning to develop CD frameworks that require few or no labeled samples.

VI. Conclusion

Due to factors such as imaging conditions, seasonal variability, and land cover changes, significant spectral differences

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often occur in unchanged areas of multitemporal remote sensing images, leading to FPs and omissions in CD. Compared with the unpredictability of changes between multitemporal images, the spectral differences in unchanged areas exhibit certain regularity. Furthermore, semi-supervised learning can simultaneously exploit the potential data distribution relationships among large volumes of unlabeled multitemporal images and derive reliable supervision signals from labeled samples. This strategy combines the strengths of both supervised and unsupervised learning, allowing for more comprehensive use of data, which enhances the accuracy and robustness of CD.

Based on these insights, we propose a distinct reconstruction-guided approach for semi-supervised HSI-CD, termed CSR-Net. This model aims to enhance CD performance by reconstructing corrected spectral sequences to mitigate spectral differences between unchanged areas across multitemporal HSIs. CSR-Net includes two specially designed subnetworks and a semi-supervised HSI-CD loss. The multitemporal spectral sequence reconstruction network encodes and reconstructs HSIs to generate corresponding spectral sequences. Then, the spectral sequence change analysis network analyzes the reconstructed sequences to detect changes. During training, the semi-supervised HSI-CD loss extracts effective supervisory information from abundant unlabeled samples and limited labeled samples to optimize the model for learning spectral–temporal relationships between multitemporal HSIs.

To validate the effectiveness and advanced nature of the proposed semi-supervised CD framework and loss function, this article conducted training, validation, and testing on four public HSI-CD datasets. Extensive experiments demonstrate that CSR-Net outperforms other state-of-the-art methods, requiring only 100 labeled samples per class. On the Farmland dataset, the Kappa coefficient reaches 94.34%.

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